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LI

$\Rightarrow d \cdot p.$

→ SVD tells you

$$A = U \Sigma V^T \rightarrow d \times d$$

$\downarrow$ $\downarrow$ $\downarrow$
 $m \times d$ $m \times m$ $m \times d$

u & v are orthogonal

Σ is pseudo-diagonal matrix.

$$m \times d \quad \begin{bmatrix} 1 & \dots & d \\ \vdots & & \\ m \end{bmatrix} = \begin{bmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_d \\ | & | & \dots & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_2 & \\ 0 & & \ddots \\ & & 0 & \ddots \end{bmatrix} \begin{bmatrix} -u_1 \\ -u_2 \\ \vdots \\ \vdots \end{bmatrix}$$

This operates on d -dimensional space.

- PCA (principal component analysis)
↳ does dimensionality reducⁿ

$A \in \mathbb{R}^{m \times d} \Rightarrow$ build a $d \times d$ covariance matrix Σ

$$C = \sum_{i=1}^n \alpha_i a_i^T$$

we check the eigen value (impt. features) and retain only the impt. features.

$$a_i \in \mathbb{R}^m \rightarrow [a_i^T w_1, a_i^T w_2, \dots, a_i^T w_k] \in \mathbb{R}^k$$

$$A = U \Sigma V^T, \quad A^T A = \underbrace{(V \Sigma U^T)(U \Sigma V^T)}_{= V \Sigma^2 V^T}$$

↓ right singular vectors.

- In general,

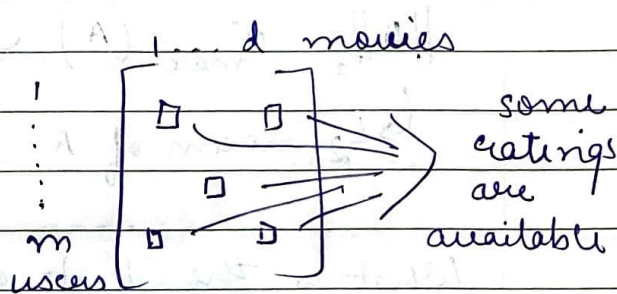
$$\left. \begin{array}{ll} A \in \mathbb{R}^{m \times d} & m > d \Rightarrow O(md^2) \\ & m \approx d \Rightarrow O(m^3) \end{array} \right\} \text{time for SVD}$$

But at scale, sometimes md^2 , is also a huge computation.

- Some common eg.

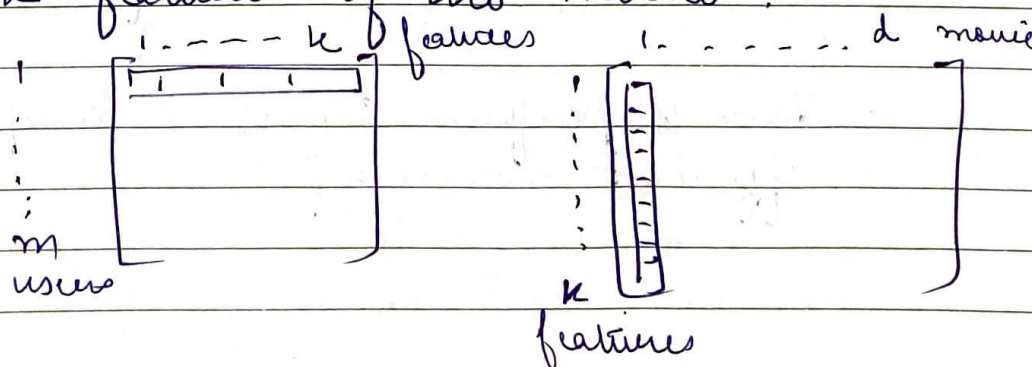
1. recommendation sys.

(user ~ movies)



every user watches at least some movies. Can we predict other values of this matrix? This is a matrix completion problem. We might have to assume few things in here.

Users will rate the movie on the basis of some k features of the movie.



low rank assumption using SVD.

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$$A_{m \times d} = B_{m \times k} C_{k \times d}$$

$$k \ll d$$

there are essentially k independent linear vectors.

- We want to faster than md^2 .

L2

From Naive SVD to Fast SVD

- Some facts about SVD -

$$A_{m \times d} = U \Sigma V^T = \sum_{i=1}^d \sigma_i \underbrace{u_i v_i^T}_{\text{rank-1 matrix}}$$

Singular values

$$\|A\|_2 = \sigma_{\max}(A) \rightarrow \text{largest singular value of given matrix}$$

$\hookrightarrow L_2$ norm of A .

What is the best rank- k approximation of A ?

$$\arg \min_{\substack{B \\ \text{rank}(B)=k}} \|A - B\|_2 = \arg \min_{\substack{B \\ \text{rank}(B)=k}} \sigma_{\max}(A - B)$$

$$\text{If } A = \sum_{i=1}^d \sigma_i u_i v_i^T \quad \text{then } B^* = \sum_{i=1}^k \sigma_i u_i v_i^T$$

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d$$

- How to perform SVD in a computationally efficient manner?

$$\{u_1, \dots, u_m\} \quad \{v_1, \dots, v_d\}$$

Let's say we are given $\{u_1, \dots, u_k\}$

orthonormal vectors

$$A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_d \\ | & | & & | \end{bmatrix}$$

$$Q = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_k \\ | & | & & | \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_k \\ | & | & & | \end{bmatrix}} \right\} \begin{array}{l} \text{this given} \\ \text{to me.} \end{array}$$

Naive SVD -

$$A = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_d \\ | & | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_d & \end{bmatrix} \begin{bmatrix} | \\ u_1 \\ | \\ u_2 \\ | \\ \vdots \\ | \\ u_d \end{bmatrix}$$

$$B = \begin{bmatrix} | & & & | \\ u_1 & \dots & & u_k \\ | & & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_k & \end{bmatrix} \begin{bmatrix} | \\ u_1 \\ | \\ \vdots \\ | \\ u_k \end{bmatrix}$$

Now, what happens in Fast SVD (given Q)

→ Compute $Q^T A = Y$ $Y \in \mathbb{R}^{k \times d}$

$$= \begin{bmatrix} | & | & & | \\ u_1 & & & \\ | & & & \\ u_2 & & & \\ | & & & \\ \vdots & & & \\ | & & & \\ u_k & & & \end{bmatrix} \begin{bmatrix} A \\ \vdots \end{bmatrix} \begin{matrix} k \times m \\ m \times d \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 \\ & & \ddots & \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_d & \end{bmatrix} \begin{bmatrix} | \\ u_1 \\ | \\ u_2 \\ | \\ \vdots \\ | \\ u_d \end{bmatrix}$$

$$Y = \begin{matrix} \sim & \sim & \sim \\ U & \Sigma & V^T \\ k \times d & k \times k & k \times d \end{matrix} \begin{matrix} \nearrow \\ \text{will} \\ \text{be same } U \end{matrix}$$

$[I]$

$$Y = \tilde{U} \tilde{\Sigma} \tilde{V}^T \quad \left(\begin{array}{l} \text{Time} \\ \text{complexity} \end{array} \quad O(dk^2) \right)$$

To get the best rank- k approximation to A ,

$$\begin{aligned} \text{set } B &= (Q \tilde{U}) \tilde{\Sigma} \tilde{V}^T \\ &= \begin{bmatrix} u_1 & \dots & u_k \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix} \begin{bmatrix} -u_1^T \\ \vdots \\ -u_k^T \end{bmatrix} \end{aligned}$$

$$A \xrightarrow{\text{SVD here}} Q(Q^T A)$$

$QQ^T \Rightarrow$ is the projection matrix onto the subspace spanned by Q .

$$\|A - QQ^T A\|_2 \leq \|A - KA\|_2$$

$$\left\| \sum_{i=1}^d \sigma_i u_i u_i^T - \sum_{i=1}^k \sigma_i u_i u_i^T \right\|_2 = \left\| \sum_{i=k+1}^d \sigma_i u_i u_i^T \right\|_2 = \sigma_{k+1}$$

We have to just make sure that,

$$\text{span}(Q) = \text{span}(\{u_1, \dots, u_k\})$$

- We can do SVD quickly, just that we should not lose info. of top- k features.

L3 Randomised SVD: Part-1

$$A \in \mathbb{R}^{m \times d} \Rightarrow A = U \Sigma V^T$$

$m \times m \quad m \times d \quad d \times d$

$$U = \begin{bmatrix} | & & | \\ u_1 & \dots & u_m \\ | & & | \end{bmatrix} \quad \text{but we want top-}k$$

$$A \approx U_k \Sigma_k V_k^T$$

$$\{u_1, \dots, u_k\}$$

$$B = Q^T A$$

$$\text{SVD}(B) = \tilde{U} \tilde{\Sigma} \tilde{V}^T$$

$$Q = \begin{bmatrix} | & & | \\ u_1 & \dots & u_k \\ | & & | \end{bmatrix}$$

$$\hat{U} = Q \tilde{U} \quad \hat{\Sigma} = \tilde{\Sigma} \quad \hat{V} = \tilde{V}$$

$$\hat{A} = \hat{U} \hat{\Sigma} \hat{V}^T$$

- Now, let's look into randomised SVD.

$$A = \begin{bmatrix} | & & | \\ a_1 & \dots & a_d \\ | & & | \end{bmatrix}_{m \times d} \quad \text{I have preserved the column space or the span of these vectors.}$$

$$\omega \sim N(\vec{0}, I)$$

$$\begin{bmatrix} \omega_1 \\ \vdots \\ \omega_d \end{bmatrix} \quad \omega_i \sim N(0, 1)$$

$$A\omega = \sum_{i=1}^d \omega_i a_i$$

$$E[A\omega] = A E[\omega] = \vec{0}$$

it has mean = $\vec{0}$

$$\begin{aligned} \text{cov}[A\omega] &= EA\omega^T \\ &= A E[\omega\omega^T] A^T \\ &= A I A^T \end{aligned}$$

$$\text{cov} = AA^T$$

and also, Gaussian. $\text{cov}[A\omega] = AA^T$

$$\Omega = \begin{bmatrix} | & | & & | \\ \omega_1 & \omega_2 & \dots & \omega_l \\ | & | & & | \end{bmatrix}$$

$O(dl)$ ←

$$\omega_i = N(\vec{0}, I)$$

$$l \ll m, d$$

this will provide the randomness

① $Y = A \Omega$ (randomly linearly combining with ω_i)

$O(mdl)$ ← $Y = \begin{bmatrix} | & & | \\ A\omega_1 & \dots & A\omega_l \\ | & & | \end{bmatrix}$

← sketch of A
picking some representative vectors of A, which should be enough.

② find the orthonormal basis of these vectors.

Gram-Schmidt's process (QR decomposition)

$O(ml^2)$ ← $Q = \text{orth}(Y)$ → $m \times l$

$O(mdl)$ ← ③ $B = Q^T A$

$O(dl^2)$ ← ④ $SVD(B) \Rightarrow B = \tilde{U} \tilde{\Sigma} \tilde{V}^T$

⑤ $\hat{U} = Q \tilde{U}, \hat{\Sigma} = \tilde{\Sigma}, \hat{V}^T = \tilde{V}^T$

$O(mlk)$ $\boxed{A = \hat{U} \hat{\Sigma} \hat{V}^T}$

Total time: $O(dl) + O(mdl) + O(dl^2) + O(mdl) + O(dl^2) + O(mlk)$

$\boxed{= O(mdl)}$ since $\Rightarrow O(mdl^2)$

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Randomised SVD - Part-2

- For exact SVD

$$\|A - QQ^T A\|_2 = \sigma_{k+1}(A) \text{ (derived previously)}$$

$Q = U_k$

$$= \begin{bmatrix} | & & | \\ u_1 & \dots & u_k \\ | & & | \end{bmatrix}$$

$\sum_{i=k+1}^d \sigma_i u_i u_i^T$

- How good is our projection matrix?

For randomised SVD,
with probability at least $1 - \delta$

$$\|A - QQ^T A\|_2 \leq \sigma_{k+1} (1 + \epsilon)$$

$$\epsilon = 6 \sqrt{\frac{k+p}{p} \ln\left(\frac{2}{\delta}\right)}$$

$p \rightarrow$ over-sampling parameter
 $l \rightarrow k+p$ (no. of linear combinations)

$$\epsilon = 1, \|A - QQ^T A\|_2 \leq 2\sigma_{k+1}$$

if this σ_{k+1} is small, then it is fine,
otherwise issues. Apriori, we don't know what is
the k value.

- By the random matrix theory arguments -

$\rightarrow$ this answers how good is QQ^T w.r.t $U_k U_k^T$

$$\|QQ^T - U_k U_k^T\|_2 \lesssim \frac{\sigma_{k+1}}{\sigma_k}$$

L5 Randomised SVD - Part 3

- $A = U \Sigma V^T$

→ we are approximating basis vectors for top- k features.

$$AA^T = (U \Sigma V^T)(U \Sigma V^T)^T$$

$$= U \Sigma V^T V \Sigma^T U^T$$

$$= U \Sigma^2 U^T$$

$$(AA^T)A = (U \Sigma^2 U^T)(U \Sigma V^T)$$

$$= \boxed{U \Sigma^3} \boxed{V^T}$$

↓ same ↓ same

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d$$

$$\sigma_1^3 \geq \sigma_2^3 \geq \dots \geq \sigma_d^3$$

→ only the singular values have changed.

$$Y = (AA^T)A \Rightarrow \text{orth}(Y) \quad \text{usual steps}$$

$$Q = (\text{orth}(Y))$$

$$\begin{aligned} \|QQ^T - U_k U_k^T\| &\leq \frac{\sigma_{k+1}((AA^T)A)}{\sigma_k((AA^T)A)} \\ &\leq \left(\frac{\sigma_{k+1}(A)}{\sigma_k(A)} \right)^3 \end{aligned}$$

- this will do much better approximation than the previous one.
why stop here?

$$(AA^T)^2(A) = U \Sigma^{2q+1} V^T$$

just by using powers of same matrix,
we are getting better results.

problem here

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$$C = (AA^T)^{\frac{1}{2}} A \quad O(md^2)$$

$$Y = C \Omega$$

$$Q = \text{orth}(Y)$$

$$B = Q^T A$$

$$\text{SVD}(B) = \tilde{U} \tilde{\Sigma} \tilde{V}^T$$

Power

iteration

$$\hat{U} = Q \tilde{U}, \quad \hat{\Sigma} = \tilde{\Sigma}, \quad \hat{V} = \tilde{V}$$

$$\hat{A} = \hat{U} \hat{\Sigma} \hat{V}^T$$

Final Algo. $\rightarrow$ Randomized SVD with Power Iteration

$$A \in \mathbb{R}^{m \times d}$$

$$Y = A \Omega \rightarrow O(mdl)$$

power iteration $O(mdl \log q)$

$$\begin{cases} \text{For } i = 1 \dots q \\ Q = \text{orth}(Y) \end{cases} \rightarrow \begin{matrix} (d \times l) \\ Z = \cancel{Q^T A} A^T Q \rightarrow O(mdl) \\ Y = A Z \rightarrow O(mdl) \\ (m \times l) \end{matrix}$$

$$B = Q^T A$$

$$\text{SVD}(B) \Rightarrow B = \tilde{U} \tilde{\Sigma} \tilde{V}^T$$

$$\hat{U} = Q \tilde{U}$$

$$\boxed{\hat{A} = \hat{U} \hat{\Sigma} \hat{V}^T}$$

$$\|QQ^T - U_k U_k^T\|_2 \leq \left(\frac{\sigma_{k+1}(A)}{\sigma_k(A)} \right)^{2q+1}$$

$$\|A - QQ^T A\|_2 \leq \left(1 + \varepsilon^{\frac{1}{2q+1}} \right) \sigma_{k+1}(A)$$